Chapter 16

Excretory Products and their Elimination

Animals pile up ammonia, urea, uric acid, carbon dioxide, water and ions such as Na+, K+, Cl–, phosphate and sulphate, either through metabolic activity or by other means such as excess ingestion, and all of it has to go, wholly or in part.

The three nitrogenous wastes

- ammonia, urea and uric acid are the major forms of nitrogenous waste that animals excrete
- ammonia is the most toxic form and calls for a large amount of water to eliminate it
- uric acid, the least toxic, can be removed with a minimum loss of water

Ammonotelism

- the process of excreting ammonia, seen in many bony fishes, aquatic amphibians and aquatic insects
- ammonia is readily soluble, so it is generally excreted as ammonium ions by diffusion across body surfaces, or through gill surfaces in fish
- the kidneys play no significant role in its removal

Ureotelism and uricotelism

- life on land made less toxic wastes such as urea and uric acid necessary, for conservation of water
- mammals, many terrestrial amphibians and marine fishes mainly excrete urea and are called ureotelic, the ammonia of metabolism being converted to urea in the liver and released into the blood, which the kidneys filter and excrete
- some urea may be held back in the kidney matrix of these animals to keep a desired osmolarity
- reptiles, birds, land snails and insects excrete uric acid as a pellet or paste with a minimum loss of water and are called uricotelic

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Kidneys, the paired bean shaped organs

Ureters, one leaving each kidney at the hilum

Urinary bladder, where the urine collects

Urethra, the outlet from the bladder

Figure 16.A Human Urinary system

16.1 Excretory structures across animals

Simple tubes to complex organs

- a survey of the animal kingdom presents a variety of excretory structures
- in most invertebrates these are simple tubular forms, whereas vertebrates have complex tubular organs, the kidneys
- protonephridia or flame cells are the excretory structures of Platyhelminthes such as Planaria, of rotifers, of some annelids and of the cephalochordate Amphioxus, and they are primarily concerned with ionic and fluid volume regulation, that is osmoregulation

Nephridia, tubules and glands

- nephridia are the tubular excretory structures of earthworms and other annelids, and they help remove nitrogenous wastes and maintain a fluid and ionic balance
- malpighian tubules are the excretory structures of most insects, cockroaches among them, and they help in the removal of nitrogenous wastes and in osmoregulation
- antennal glands or green glands perform the excretory function in crustaceans such as prawns

16.2 The human excretory system

The organs, and the kidney outside

- in humans the excretory system consists of a pair of kidneys, one pair of ureters, a urinary bladder and a urethra
- the kidneys are reddish brown, bean shaped structures lying between the levels of the last thoracic and the third lumbar vertebra, close to the dorsal inner wall of the abdominal cavity
- each kidney of an adult measures 10-12 cm in length, 5-7 cm in width and 2-3 cm in thickness, with an average weight of 120-170 g
- a notch called the hilum, towards the centre of the inner concave surface, admits the ureter, the blood vessels and the nerves, and inner to it lies a broad funnel shaped space, the renal pelvis, with projections called calyces

The kidney inside

- the outer layer of the kidney is a tough capsule
- within it lie two zones, an outer cortex and an inner medulla
- the medulla is divided into a few conical masses, the medullary pyramids, that project into the calyces, the singular being calyx

Capsule, the tough outer layer

Cortex outside and medulla within

Medullary pyramids, projecting into the calyces

Renal columns, the cortex reaching between the pyramids

Renal pelvis, with the ureter leaving at the hilum

Figure 16.B Longitudinal section (Diagrammatic) of Kidney

Afferent arteriole in, efferent arteriole out

Glomerulus, the tuft of capillaries

Bowman's capsule around it

Renal tubule, leading on to the collecting duct

Figure 16.C A diagrammatic representation of a nephron showing blood vessels, duct and tubule

16.3 The nephron

Renal columns

- the cortex extends in between the medullary pyramids as renal columns
- these columns are called the Columns of Bertini

A million functional units

- each kidney has nearly one million complex tubular structures called nephrons, which are its functional units
- each nephron has two parts, the glomerulus and the renal tubule

The glomerulus

- the glomerulus is a tuft of capillaries formed by the afferent arteriole, a fine branch of the renal artery
- blood is carried away from the glomerulus by an efferent arteriole

Where the tubule begins

- the renal tubule begins with a double walled cup like structure, the Bowman's capsule, which encloses the glomerulus
- the glomerulus together with the Bowman's capsule is called the malpighian body or renal corpuscle
- the tubule continues to form a highly coiled network, the proximal convoluted tubule or PCT

Afferent arteriole entering, efferent arteriole leaving

Bowman's capsule, the double walled cup

Glomerulus enclosed within it

Proximal convoluted tubule leading away

Figure 16.D Malpighian body (renal corpuscle)

16.4 Parts of the renal tubule

From the PCT to the collecting duct

- a hairpin shaped Henle's loop is the next part of the tubule, with a descending and an ascending limb
- the ascending limb continues as another highly coiled region, the distal convoluted tubule or DCT
- the DCTs of many nephrons open into a straight tube called the collecting duct, and many of these converge and open into the renal pelvis through the medullary pyramids in the calyces

Cortical and juxta medullary nephrons

- the malpighian corpuscle, the PCT and the DCT sit in the cortical region of the kidney, while the loop of Henle dips into the medulla
- in the majority of nephrons the loop of Henle is too short and reaches only a very little way into the medulla, and these are called cortical nephrons
- in some nephrons the loop of Henle is very long and runs deep into the medulla, and these are called juxta medullary nephrons

Peritubular capillaries and vasa recta

- the efferent arteriole emerging from the glomerulus forms a fine capillary network around the renal tubule, the peritubular capillaries
- a minute vessel of this network runs parallel to the Henle's loop, forming a U shaped vasa recta
- vasa recta is absent or highly reduced in cortical nephrons

16.5 Urine formation

Glomerular filtration

- urine formation involves three main processes, glomerular filtration, reabsorption and secretion, each taking place in different parts of the nephron
- the first step is the filtration of blood by the glomerulus, and on an average 1100-1200 ml of blood is filtered by the kidneys per minute, roughly 1/5th of the blood pumped out by each ventricle of the heart in a minute
- the glomerular capillary blood pressure drives blood through 3 layers, the endothelium of the glomerular blood vessels, the epithelium of the Bowman's capsule and a basement membrane between the two
- the epithelial cells of the Bowman's capsule, the podocytes, are arranged so intricately as to leave minute spaces called filtration slits or slit pores, and the blood is filtered so finely that almost every constituent of the plasma other than the proteins passes into the lumen of the capsule, which is why this is considered a process of ultra filtration

16.6 Glomerular filtration rate

What GFR is, and how it is held steady

- the amount of filtrate formed by the kidneys per minute is the glomerular filtration rate or GFR, approximately 125 ml per minute in a healthy individual, that is 180 litres per day
- the kidneys have built-in mechanisms to regulate it, one efficient mechanism being the juxta glomerular apparatus or JGA
- the JGA is a special sensitive region formed by cellular modifications in the distal convoluted tubule and the afferent arteriole where the two touch
- a fall in GFR can activate the JG cells to release renin, which can stimulate the glomerular blood flow and so bring the GFR back to normal

Reabsorption and secretion

- comparing the filtrate formed per day, 180 litres, with the urine released, 1.5 litres, shows that nearly 99 per cent of the filtrate has to be taken back by the renal tubules, which is reabsorption
- the tubular epithelial cells of the different segments do this either actively or passively, so that glucose, amino acids, Na+ and the like are reabsorbed actively while nitrogenous wastes are absorbed by passive transport, and water is reabsorbed passively in the initial segments
- during urine formation the tubular cells secrete substances such as H+, K+ and ammonia into the filtrate
- tubular secretion matters because it helps maintain the ionic and acid base balance of the body fluids

16.7 Function of the tubules

Proximal convoluted tubule (PCT)

- lined by a simple cuboidal brush border epithelium, which increases the surface area for reabsorption
- nearly all the essential nutrients and 70-80 per cent of the electrolytes and water are reabsorbed by this segment
- it also helps maintain the pH and the ionic balance of the body fluids, by selective secretion of hydrogen ions and ammonia into the filtrate and by absorption of HCO_3^- from it

Henle's loop and the DCT

- reabsorption is minimum in the ascending limb, yet this region matters greatly in maintaining the high osmolarity of the medullary interstitial fluid
- the descending limb is permeable to water but almost impermeable to electrolytes, which concentrates the filtrate as it moves down, while the ascending limb is impermeable to water but lets electrolytes pass actively or passively, so the concentrated filtrate is diluted as it moves up and electrolytes pass to the medullary fluid
- conditional reabsorption of Na+ and water takes place in the DCT, which is also capable of reabsorbing HCO_3^- and of selectively secreting hydrogen and potassium ions and NH_3 to maintain the pH and the sodium-potassium balance in blood

PCT, where nutrients, electrolytes and water are taken back

Descending limb, permeable to water

Ascending limb, moving electrolytes out

DCT and collecting duct, secreting H+ and K+ and taking back water

Figure 16.E Reabsorption and secretion of major substances at different parts of the nephron (Arrows indicate direction of movement of materials.)

16.8 The collecting duct

A long duct across both zones

- this long duct extends from the cortex of the kidney to the inner parts of the medulla
- large amounts of water can be reabsorbed here to produce a concentrated urine
- the segment allows small amounts of urea to pass into the medullary interstitium, keeping up the osmolarity
- it also helps maintain the pH and ionic balance of blood, by the selective secretion of H+ and K+ ions

16.9 Concentrating the filtrate

Counter current flow

- mammals have the ability to produce a concentrated urine, and the Henle's loop and the vasa recta matter greatly in this
- the flow of filtrate in the two limbs of Henle's loop runs in opposite directions and so forms a counter current
- the flow of blood through the two limbs of the vasa recta follows a counter current pattern as well

Henle's loop, descending and ascending limbs side by side

Vasa recta, running parallel to the loop

Cortex at 300 mOsmolL–1, inner medulla at about 1200 mOsmolL–1

NaCl and urea, moving between limb and interstitium

Figure 16.F Diagrammatic representation of a nephron and vasa recta showing counter current mechanisms

16.10 The counter current mechanism

The gradient it builds

- the closeness of the Henle's loop and the vasa recta, together with the counter current in them, helps maintain an osmolarity that rises towards the inner medullary interstitium, from 300 mOsmolL–1 in the cortex to about 1200 mOsmolL–1 in the inner medulla
- the gradient is caused mainly by NaCl and urea

How the two solutes travel

- NaCl is transported by the ascending limb of Henle's loop and exchanged with the descending limb of the vasa recta
- it is returned to the interstitium by the ascending portion of the vasa recta
- small amounts of urea likewise enter the thin segment of the ascending limb of Henle's loop and are carried back to the interstitium by the collecting tubule
- this transport of substances, made possible by the special arrangement of Henle's loop and vasa recta, is called the counter current mechanism

What the gradient is for

- the mechanism helps maintain a concentration gradient in the medullary interstitium
- such an interstitial gradient lets water pass easily out of the collecting tubule, and so concentrates the filtrate, that is the urine
- human kidneys can produce urine nearly four times as concentrated as the filtrate first formed

Osmoreceptors and ADH

- kidney function is monitored and regulated by hormonal feedback mechanisms involving the hypothalamus, the JGA and, to a certain extent, the heart
- osmoreceptors in the body are activated by changes in blood volume, body fluid volume and ionic concentration, and an excessive loss of fluid activates them, so that they stimulate the hypothalamus to release antidiuretic hormone (ADH) or vasopressin from the neurohypophysis
- ADH facilitates water reabsorption from the later parts of the tubule and so prevents diuresis, and a rise in body fluid volume switches the osmoreceptors off and suppresses ADH release, completing the feedback
- ADH also acts on the kidney by constricting blood vessels, which raises blood pressure, and a rise in blood pressure can raise the glomerular blood flow and so the GFR

16.11 Regulation of kidney function

The renin-angiotensin mechanism

- the JGA plays a complex regulatory role
- a fall in glomerular blood flow, glomerular blood pressure or GFR can activate the JG cells to release renin, which converts angiotensinogen in blood to angiotensin I and further to angiotensin II
- angiotensin II, a powerful vasoconstrictor, raises the glomerular blood pressure and so the GFR, and it also activates the adrenal cortex to release aldosterone
- aldosterone causes reabsorption of Na+ and water from the distal parts of the tubule, which again raises blood pressure and GFR, and this whole mechanism is known as the renin-angiotensin mechanism

The check on it

- an increase in blood flow to the atria of the heart can cause the release of Atrial Natriuretic Factor (ANF)
- ANF can cause vasodilation, the dilation of blood vessels, and so lower the blood pressure
- the ANF mechanism therefore acts as a check on the renin-angiotensin mechanism

16.12 Micturition

The bladder waits for a signal

- urine formed by the nephrons is finally carried to the urinary bladder, where it is stored until a voluntary signal comes from the central nervous system (CNS)
- the signal is set off by the stretching of the bladder as it fills
- the stretch receptors on the walls of the bladder send signals to the CNS in response

The micturition reflex

- the CNS passes on motor messages that start the contraction of the smooth muscles of the bladder and, at the same time, the relaxation of the urethral sphincter, releasing the urine
- the process of release is called micturition, and the neural mechanism causing it is the micturition reflex

What the urine is like

- an adult human excretes on an average 1 to 1.5 litres of urine per day
- it is a light yellow coloured watery fluid, slightly acidic at pH-6.0, with a characterestic odour, and on an average 25-30 gm of urea leaves in it per day
- various conditions can alter its characteristics, so analysis of urine helps in the clinical diagnosis of many metabolic discorders and of kidney malfunction
- glucose in urine, that is glycosuria, and ketone bodies, that is ketonuria, point to diabetes mellitus

16.13 Role of other organs in excretion

Lungs and liver

- besides the kidneys, the lungs, the liver and the skin also help eliminate excretory wastes
- the lungs remove large amounts of CO2, approximately 200mL per minute, and significant quantities of water every day
- the liver, the largest gland in the body, secretes bile containing bilirubin, biliverdin, cholesterol, degraded steroid hormones, vitamins and drugs, most of which finally pass out along with the digestive wastes

Skin, and even saliva

- the sweat and sebaceous glands in the skin can eliminate certain substances through their secretions
- sweat is a watery fluid containing NaCl and small amounts of urea, lactic acid and the like, and although its primary function is to cool the body surface, it also removes some of these wastes
- sebaceous glands eliminate sterols, hydrocarbons and waxes through sebum, a secretion that provides a protective oily covering for the skin
- small amounts of nitrogenous waste can leave through saliva too

16.14 Disorders of the excretory system

Uremia and haemodialysis

- malfunctioning kidneys let urea build up in the blood, a condition called uremia, which is highly harmful and may lead to kidney failure
- in such patients urea can be removed by haemodialysis, in which blood drained from a convenient artery is pumped, after an anticoagulant such as heparin is added, into a dialysing unit called an artificial kidney
- the unit holds a coiled cellophane tube surrounded by a dialysing fluid of the same composition as plasma except for the nitrogenous wastes, and the porous cellophane membrance lets molecules pass on a concentration gradient
- as the nitrogenous wastes are absent from the dialysing fluid they move out freely and clear the blood, which is then pumped back into the body through a vein after anti-heparin is added, a method that has been a boon for thousands of uremic patients all over the world

Transplantation, and two more disorders

- kidney transplantation is the ultimate method of correcting acute renal failure, and the functioning kidney used comes preferably from a close relative, to minimise the chances of rejection by the immune system of the host
- modern clinical procedures have raised the success rate of this complicated technique
- renal calculi are a stone or insoluble mass of crystallised salts, oxalates and the like, formed within the kidney
- glomerulonephritis is inflammation of the glomeruli of the kidney

16.15 Summary

Wastes and the organs that remove them

- many nitrogen containing substances, ions, CO_2, water and the like accumulate in the body and have to be eliminated, and the nature of the nitrogenous waste formed and the way it is excreted vary among animals, mainly with the habitat and the availability of water
- ammonia, urea and uric acid are the major nitrogenous wastes excreted
- protonephridia, nephridia, malpighian tubules, green glands and kidneys are the common excretory organs in animals, and they not only eliminate nitrogenous wastes but also help maintain the ionic and acid-base balance of the body fluids

The human plan

- in humans the excretory system consists of one pair of kidneys, a pair of ureters, a urinary bladder and a urethra
- each kidney has over a million tubular structures called nephrons, the functional unit, with two portions, the glomerulus and the renal tubule
- the glomerulus is a tuft of capillaries formed from afferent arterioles, fine branches of the renal artery, and the renal tubule starts with a double walled Bowman's capsule and is further differentiated into a PCT, a Henle's loop (HL) and a DCT
- the DCTs of many nephrons join a common collecting duct, many of which finally open into the renal pelvis through the medullary pyramids, and the Bowman's capsule encloses the glomerulus to form the malpighian or renal corpuscle

How urine is made and concentrated

- urine formation involves filtration, reabsorption and secretion, filtration being a non-selective process performed by the glomerulus with the glomerular capillary blood pressure, about 1200 ml of blood yielding 125 ml of filtrate in the Bowman's capsule per minute, which is the GFR, regulated in good part by the JGA
- nearly 99 per cent of the filtrate is reabsorbed across the different parts of the nephron, the PCT being the major site of reabsorption and selective secretion, while the HL primarily maintains the osmolar gradient of 300 mOsmolL–1 to 1200 mOsmolL–1 within the kidney interstitium
- the DCT and collecting duct allow extensive reabsorption of water and certain electrolytes, which helps in osmoregulation, and H+, K+ and NH3 can be secreted into the filtrate by the tubules to maintain the ionic balance and pH of the body fluids
- a counter current mechanism operates between the two limbs of the loop of Henle and those of the vasa recta, the capillary parallel to the loop, so the filtrate is concentrated as it moves down the descending limb and diluted by the ascending limb, electrolytes and urea being retained in the interstitium, and the DCT and collecting duct concentrate the filtrate about four times, from 300 mOsmolL–1 to 1200 mOsmolL–1, an excellent mechanism of conservation of water
- urine is stored in the urinary bladder until a voluntary signal from the CNS carries out its release through the urethra, that is micturition, and the skin, lungs and liver also assist in excretion

16.16 Exercises

From this chapter

1. Define Glomerular Filtration Rate (GFR)
2. Explain the autoregulatory mechanism of GFR.
3. Indicate whether the following statements are true or false : (a) Micturition is carried out by a reflex. (b) ADH helps in water elimination, making the urine hypotonic. (c) Protein-free fluid is filtered from blood plasma into the Bowman's capsule. (d) Henle's loop plays an important role in concentrating the urine. (e) Glucose is actively reabsorbed in the proximal convoluted tubule.
4. Give a brief account of the counter current mechanism.
5. Describe the role of liver, lungs and skin in excretion.
6. Explain micturition.
7. Match the items of column I with those of column II : Column I (a) Ammonotelism (b) Bowman's capsule (c) Micturition (d) Uricotelism (d) ADH , Column II (i) Birds (ii) Water reabsorption (iii) Bony fish (iv) Urinary bladder (v) Renal tubule

16.17 Exercises, continued

From this chapter, continued

1. What is meant by the term osmoregulation?
2. Terrestrial animals are generally either ureotelic or uricotelic, not ammonotelic, why ?
3. What is the significance of juxta glomerular apparatus (JGA) in kidney function?
4. Name the following: (a) A chordate animal having flame cells as excretory structures (b) Cortical portions projecting between the medullary pyramids in the human kidney (c) A loop of capillary running parallel to the Henle's loop.
5. Fill in the gaps : (a) Ascending limb of Henle's loop is _____ to water whereas the descending limb is _____ to it. (b) Reabsorption of water from distal parts of the tubules is facilitated by hormone _____. (c) Dialysis fluid contain all the constituents as in plasma except _____. (d) A healthy adult human excretes (on an average) _____ gm of urea/day.